

Tests on running inference on the mediapipe .tflite obd model inference tool web for l3f

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Situation overview

This document is a summary of tests run on the mediapipe .tflite obd model inference web tool I built for the l3f project. It aims to confirm the web tool is reliable & usable across multiple hardware options without crashing & misbehavior. This is a summary of multiple tests

Setup

Tests For every test we record the following environment variables:

- device
- OS
- browser + version
- camera
- target sampling rate
- model
- delegate
- score threshold
- goal of test
- duration of test
- special notes on env

, where

- **browser + version** is (index.html -> Environment -> Browser) rendering
- **target sampling rate** is the fps parameter given to `startCameraInferenceLoop()` when called from within `handleStartCamera()` in `src/main.js`
- **model** is: DIRNAME where `data_and_training/models/DIRNAME/model.tflite` is identical to `browser_app/public/models/model.tflite`; Test: `cmp browser_app/public/models/model.tflite data_and_training/models/DIRNAME/model.tflite -> silent exit, no browser_app/public/models/model.tflite data_and_training/models/DIRNAME/model.tflite differ: byte xx, line y type output.`
- **delegate** is the `delegate` parameter given to `ObjectDetector.createFromOptions` in model loading/compilation section of `try` block of `startApplication()` function in `src/main.js`
- **score threshold** is the `scoreThreshold` parameter given to `ObjectDetector.createFromOptions` in model loading/compilation section of `try` block of `startApplication()` function in `src/main.js`

and we collect

- csv formatted measurements (camera fpc, sampling fps, inference fps, inference latency) & appropriate visualisations
- UI behavior observations
- browser command line prints of note
- other notes

, where

browser command line prints of note means we ignore per-detection/per-measurement prints, but note any execution time errors.

Technical Render this document by running `Rscript -e "rmarkdown::render('reporting.Rmd')"`
From `browser_app/measurements/`

Libraries:

```
# For data manipulation
suppressPackageStartupMessages(library(dplyr)) # <- Don't print warning on import
library(tidyr)

# For plotting & rendering
library(ggplot2)
library(scales)
```

Diagram drawing function (edited for Rmd use from `data_and_training/devtools/fpsdiagram.R`):

```
fpsdiagram <- function(input_filename) {

  if (!file.exists(input_filename)) {
    stop(
      paste0("Error: File '", input_filename, "' not found."),
      call. = FALSE
    )
  }

  df <- read.csv(input_filename, stringsAsFactors = FALSE)

  df$Timestamp <- as.POSIXct(
    df$Timestamp,
    format = "%Y-%m-%dT%H:%M:%OSZ",
    tz = "UTC"
  )

  df_long <- df %>%
    pivot_longer(
      cols = c(Camera_FPS, Sampling_FPS, Inference_FPS),
      names_to = "Metric",
      values_to = "FPS"
    )

  p <- ggplot(
    df_long,
    aes(x = Timestamp, y = FPS, color = Metric)
  ) +
    geom_line(linewidth = 1) +
    geom_point(size = 2) +
    scale_x_datetime(labels = date_format("%H:%M:%S")) +
    labs(
      title = "Performance Monitoring Metrics",
      subtitle = paste("File:", input_filename),
      x = "Timestamp (UTC)",
      y = "FPS",
      color = "Metric"
    ) +
```

```

theme_minimal(base_size = 12) +
theme(
  legend.position = "bottom",
  axis.text.x = element_text(angle = 45, hjust = 1)
)

print(p)
}

```

Test series 1: Finding the optimal sampling rate

Test 1.1 starting from a low baseline

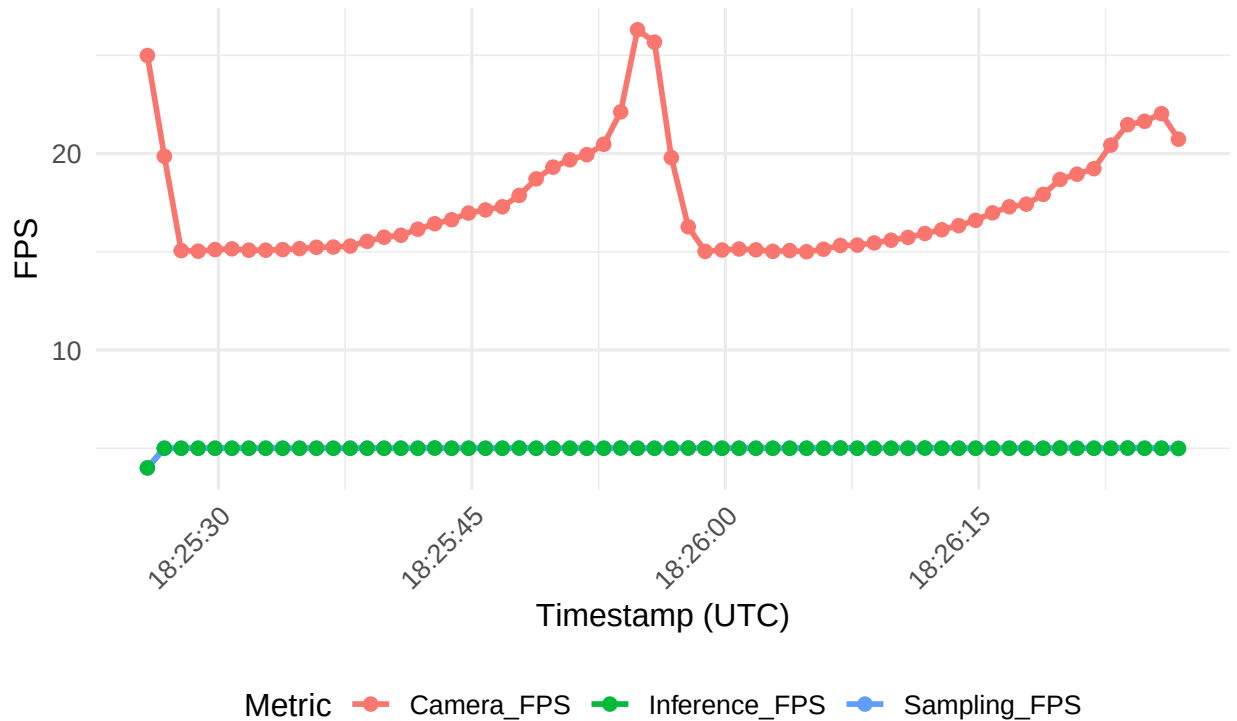
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 5
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr5_20260921_212518.csv")
```

Performance Monitoring Metrics

File: fps_sr5_20260921_212518.csv



Results

No record of browser command line prints.

Notes: Inference_FPS matches Sampling_FPS, significantly below Camera_FPS, can raise.

Test 1.2 raising to 10 fps sampling

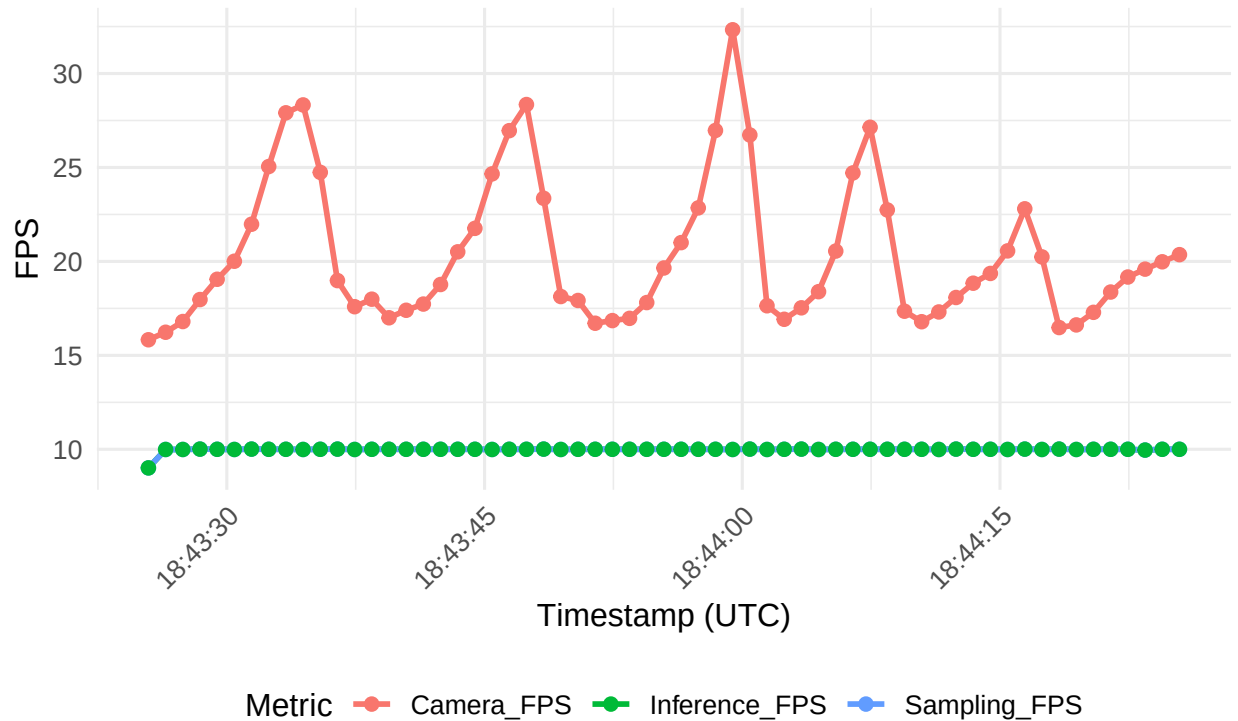
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 10
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establichment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr10_20260921_214321.csv")
```

Performance Monitoring Metrics

File: fps_sr10_20260921_214321.csv



Results

No record of browser command line prints.

Notes: Inference_FPS matches Sampling_FPS, still below Camera_FPS, can raise.

Test 1.3 reaching camera fps

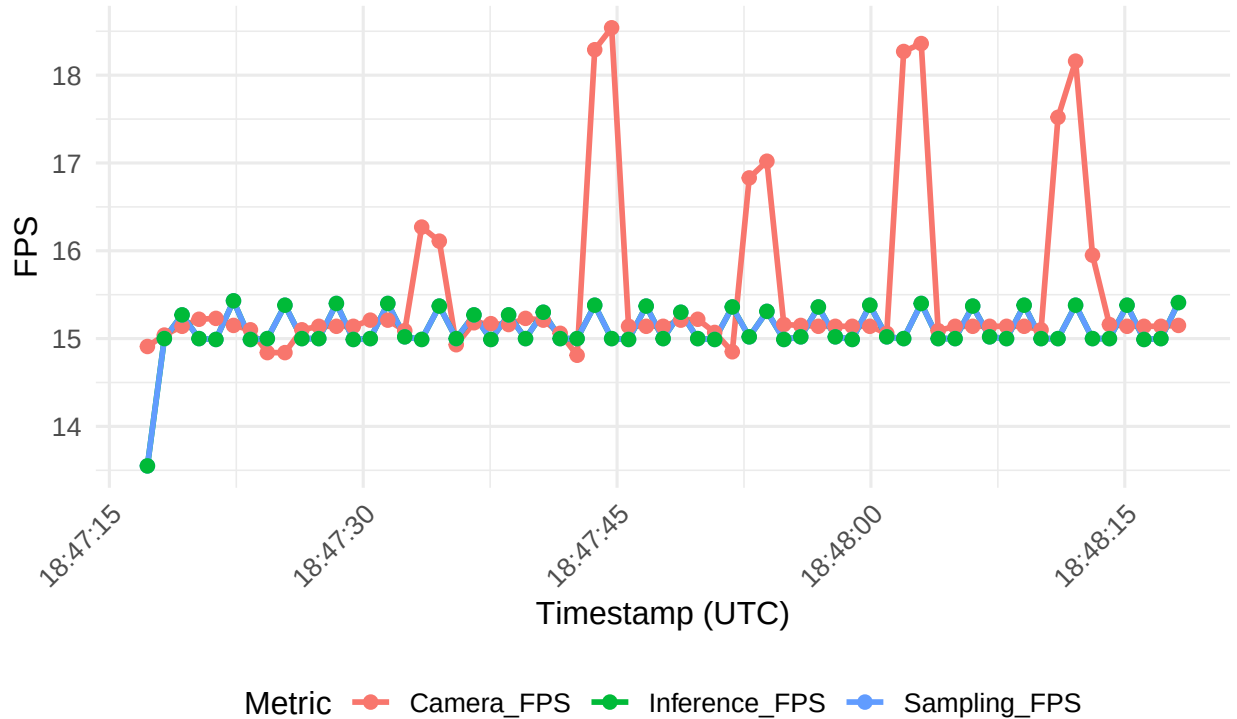
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 15
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr15_20260921_214710.csv")
```

Performance Monitoring Metrics

File: fps_sr15_20260921_214710.csv



Results

No record of browser command line prints.

Notes: Inference_FPS & Sampling_FPS now occasionally above Camera_FPS, will lower.

Test 1.4 achieving stability

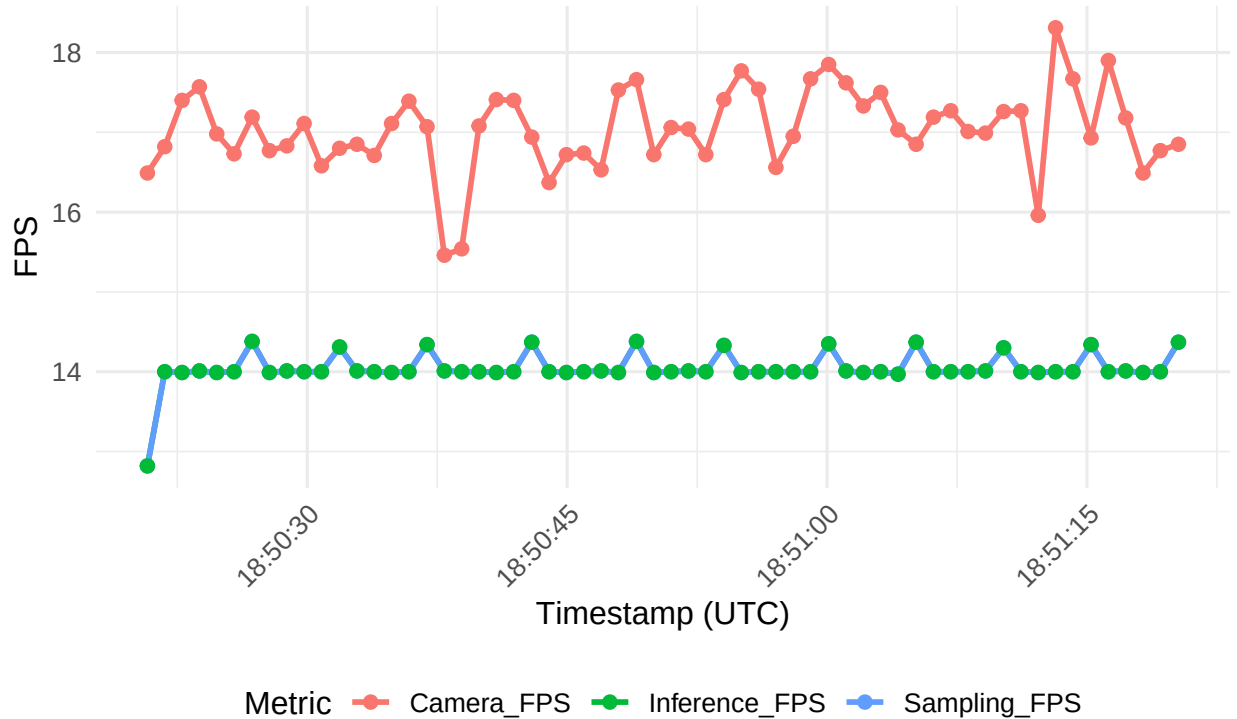
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr14_20260921_215016.csv")
```

Performance Monitoring Metrics

File: fps_sr14_20260921_215016.csv



Results

No record of browser command line prints.

Notes: Inference_FPS & Sampling_FPS now stable below Camera_FPS, not surpassing it. NOTE: Latency_MS (how long it actually takes to complete an inference) ranges from 46 to 49. It is still realistic to raise it. We're bottlenecked by Camera_FPS on this hardware.

Test series 2: On different devices

Test 2.1 on an old laptop (Thinkpad T420)

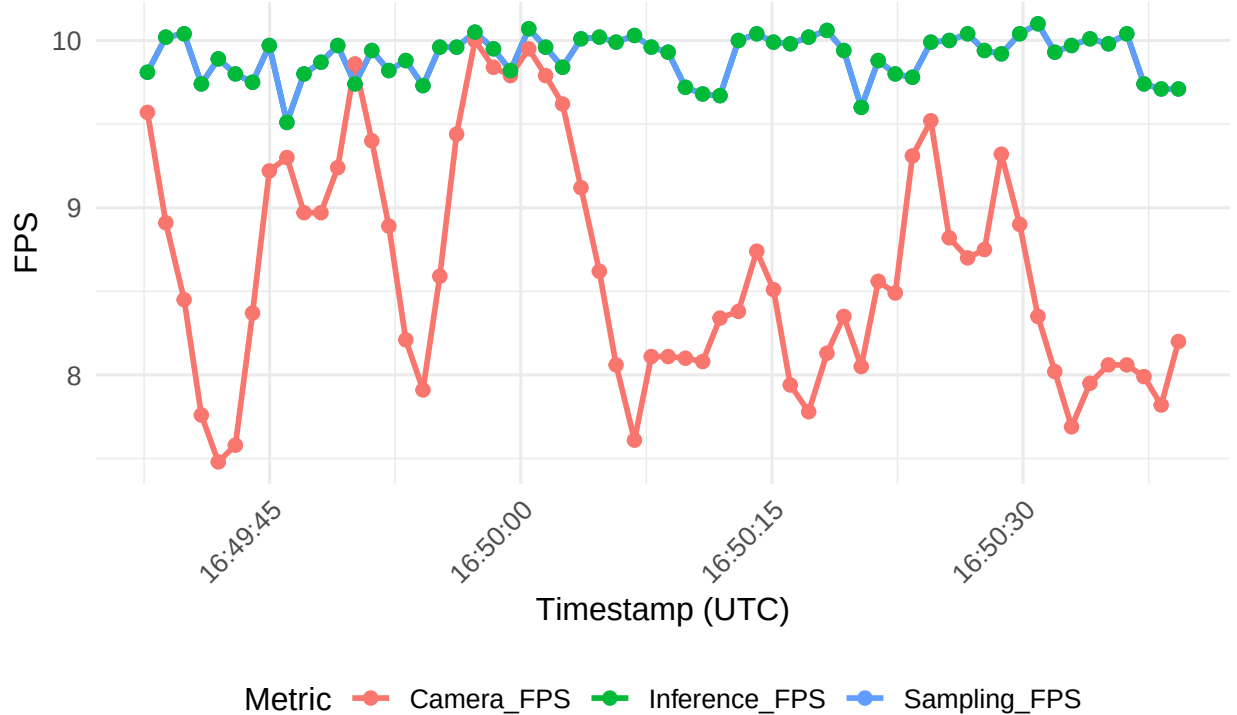
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Integrated Camera
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_l3f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: baseline on laptop
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_20260922_194937.csv")
```

Performance Monitoring Metrics

File: fps_t420_20260922_194937.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: Camera_FPS ranges from below 8 to just around 10. Sampling_FPS & Inference_FPS hovers around 10. Latency_MS ranges from 97 to 103. The limiting factor with this setup is still Camera_FPS, but it's noticeably lower than on other computer. Will set sampling rate to 7 to achieve stability.

Test 2.2 T420, lowered sample rate

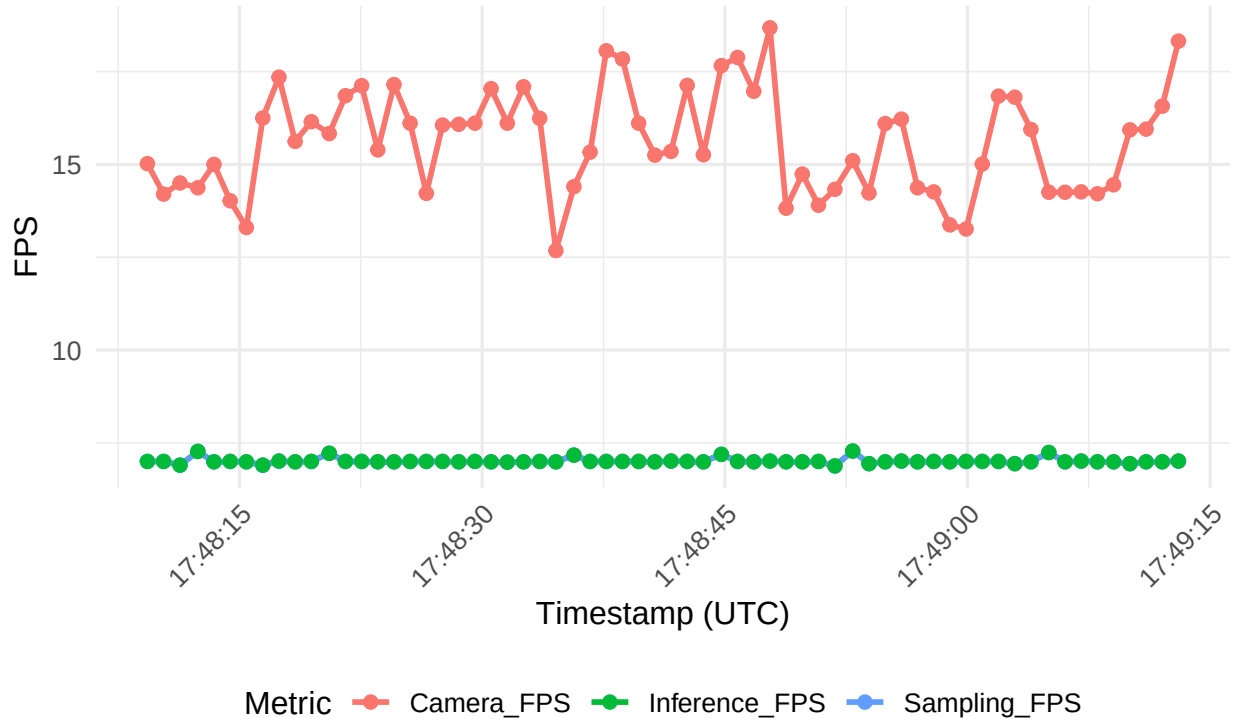
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Integrated Camera
- target sampling rate: 7
- model: ssd_plus_mobilenet_v2_13f_20260807_225848
- delegate: CPU
- score threshold: 0.35
- goal of test: baseline on laptop
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_sr7_20260922_204808.csv")
```

Performance Monitoring Metrics

File: fps_t420_sr7_20260922_204808.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: **Camera_FPS** suddenly jumped to around 15. So the limiting factor probably wasn't the **Camera_FPS** but the inference. Inferences now range from 98 to 114, which is unusual. Trying to reach a slightly better baseline at 8 fps.

Test 2.3 T420, attempting to reach a better baseline at 8fps

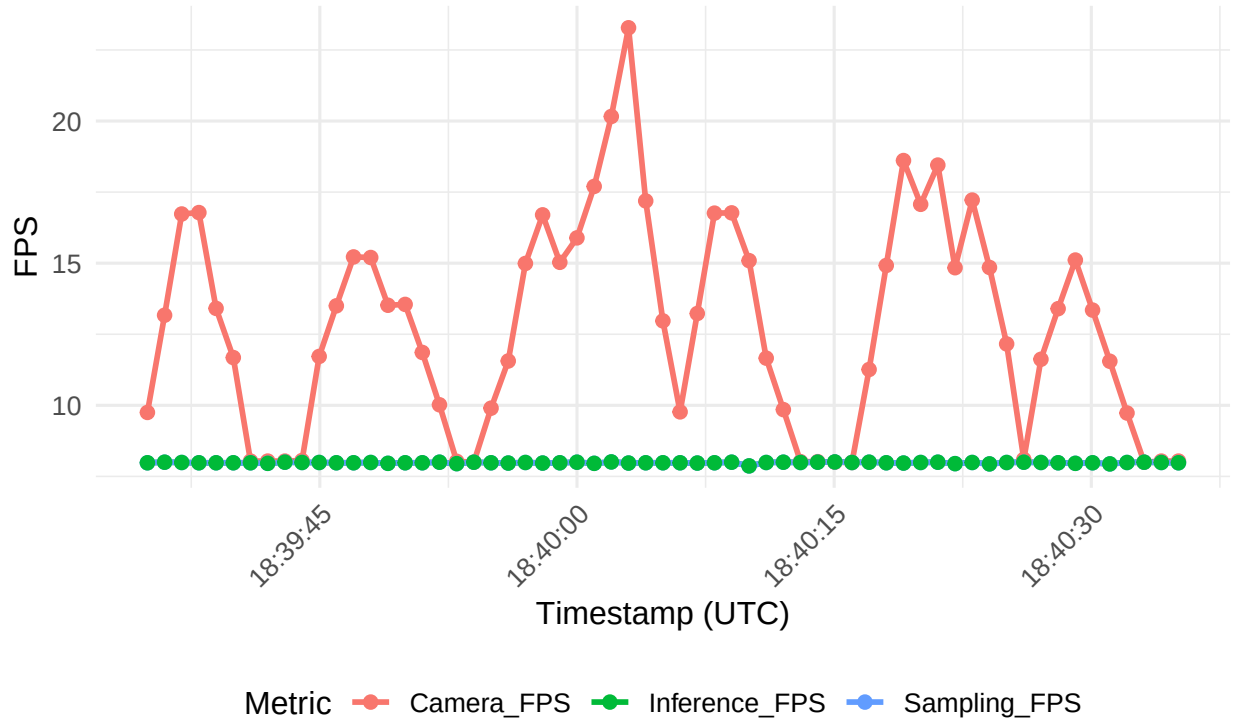
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Lubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Integrated Camera
- target sampling rate: 8
- model: `ssd_plus_mobilenet_v2_l3f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline on laptop
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_sr8_20260922_213934.csv")
```

Performance Monitoring Metrics

File: fps_t420_sr8_20260922_213934.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: **Camera_FPS** behaves in a weird & unexpected manner; Spikes reach over 20, pits are at inference/sampling fps. Rates. Inference latency range is huge (96 to 115), but a huge portion of it is around 100-105, with only a few exceptions at the tails. The results for this test show that it's better to keep using 7 fps sampling on laptop still. So that is my laptop baseline from now on. Next I will try whether or not changing from the integrated camera to an external webcam has any effect on anything. The hypothesis is no.

Test 2.4 T420, changing the camera

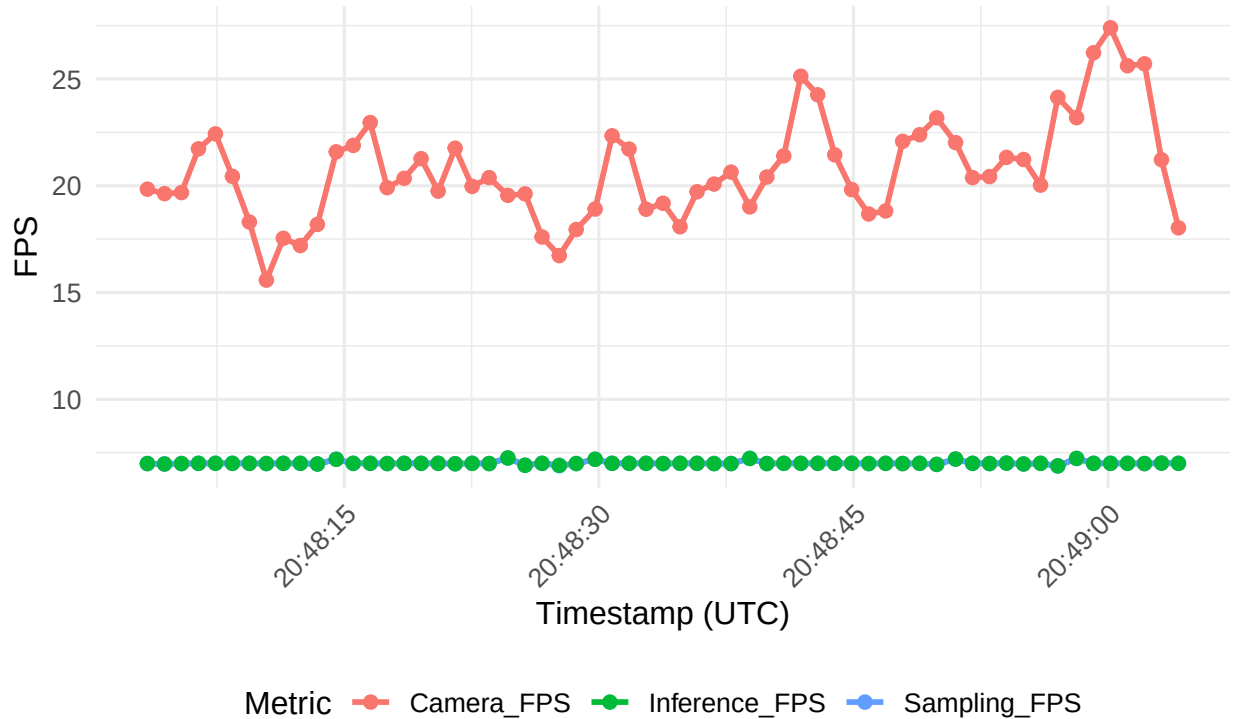
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 7
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: seeing the affect of changing the camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_base.csv")
```

Performance Monitoring Metrics

File: fps_t420_logitech_base.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

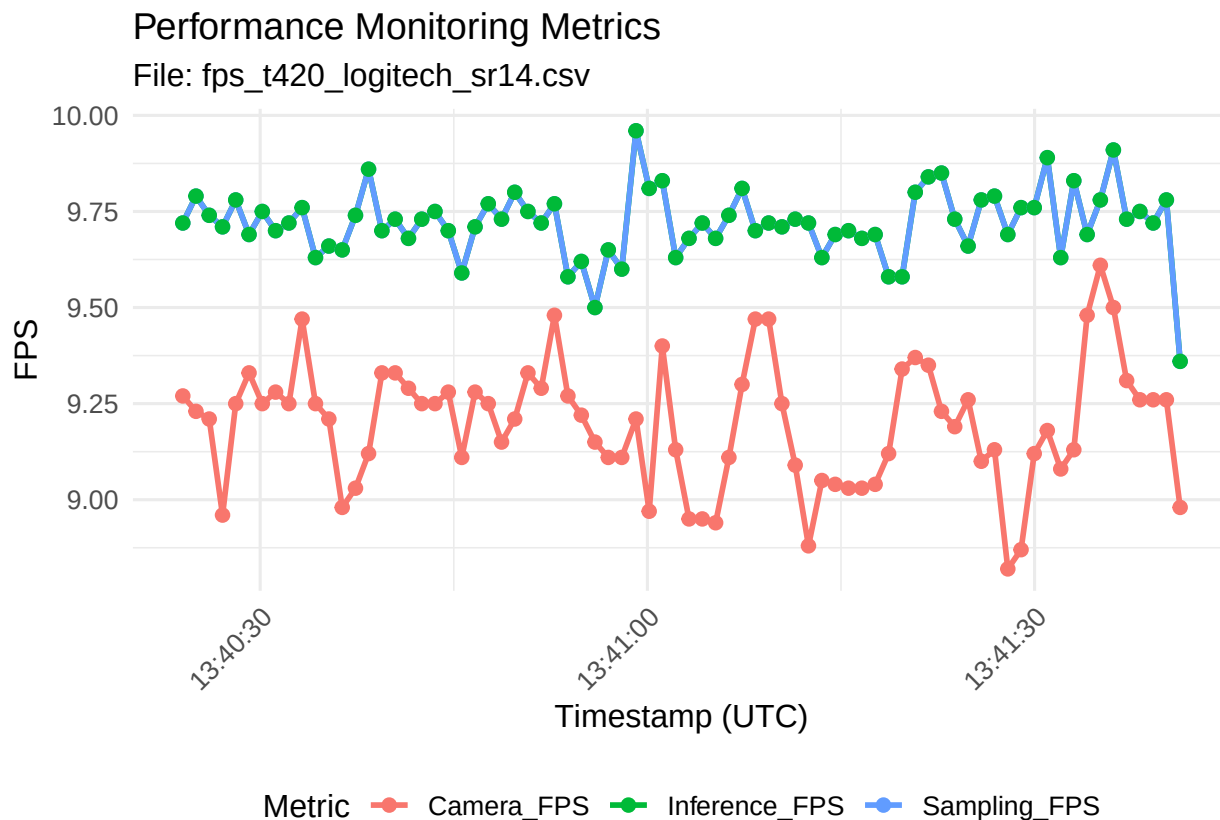
Notes: Contrary to my expectations, there is a wild jump in **Camera_FPS**. All values are now over 15 fps, meaning that sampling rate can be increased. **Latency_MS** ranges from 101 to 112. This prompts two new investigations: sr8 and sr14 on t420+logitech

Test 2.5 T420, trying higher framerate with camera

Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: attempting to establish new baseline on t420 with dedicated camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_sr14.csv")
```



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: As we expected, the hardware is unable to produce a sampling rate of 14, with **Sampling_FPS** moving around 9.75. But lowest tips are still above 9.25. Maybe it's worth to test if 9 is an acceptable sampling rate on t420+logitech, and only move test sr8 if we find 9 is not an acceptable baseline. For some reason **Camera_FPS** is below **Inference_FPS** now when we stress the system with a high sr target. The hypothesis that **Camera_FPS** is independent of the other systems seems to be wrong. **Latency_MS** ranges from 96 to 107, which would indicate an around 10 ms **Sampling_FPS**

Test 2.6 T420, attempting to establish sr9 baseline

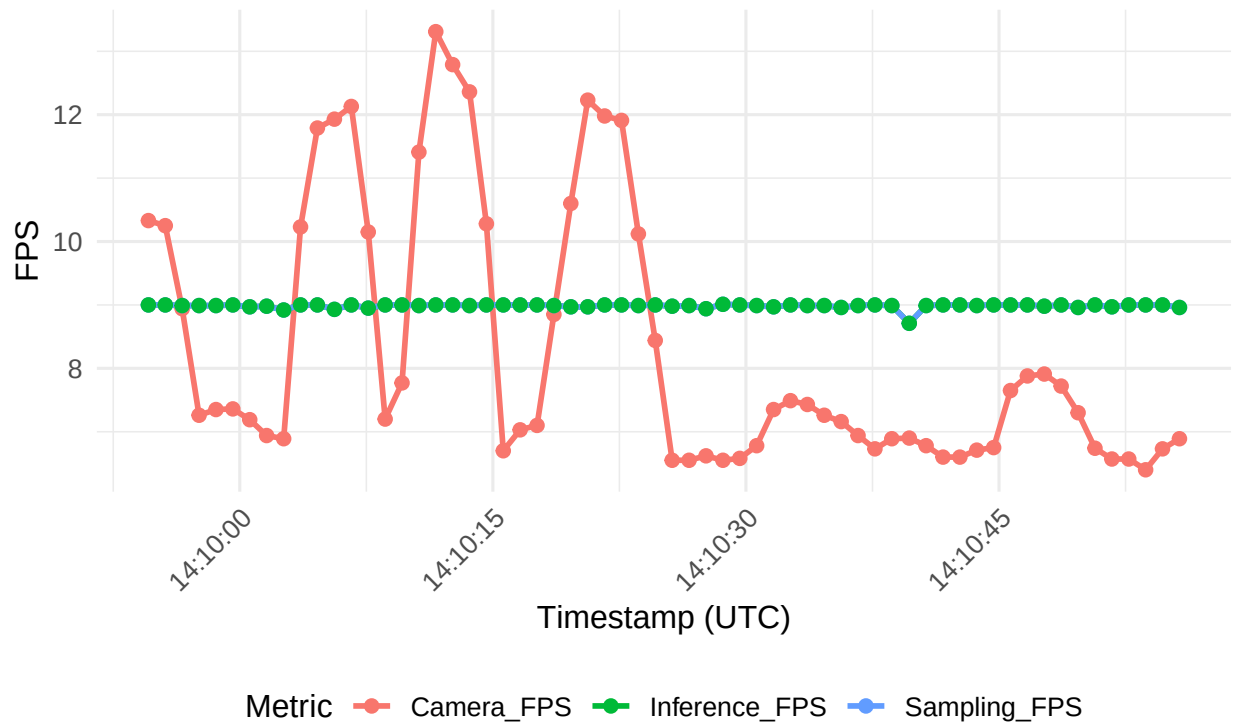
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 9
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: attempting to establish new baseline on t420 with dedicated camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_sr9.csv")
```

Performance Monitoring Metrics

File: fps_t420_logitech_sr9.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: Camera_FPS does weird fluctuation, occasionally going below Sampling_FPS. Thus sr9 is unacceptable. One more measurement on t420+logitech ought to be done with sr8.

Test 2.7 T420, attempting to establish sr8 baseline

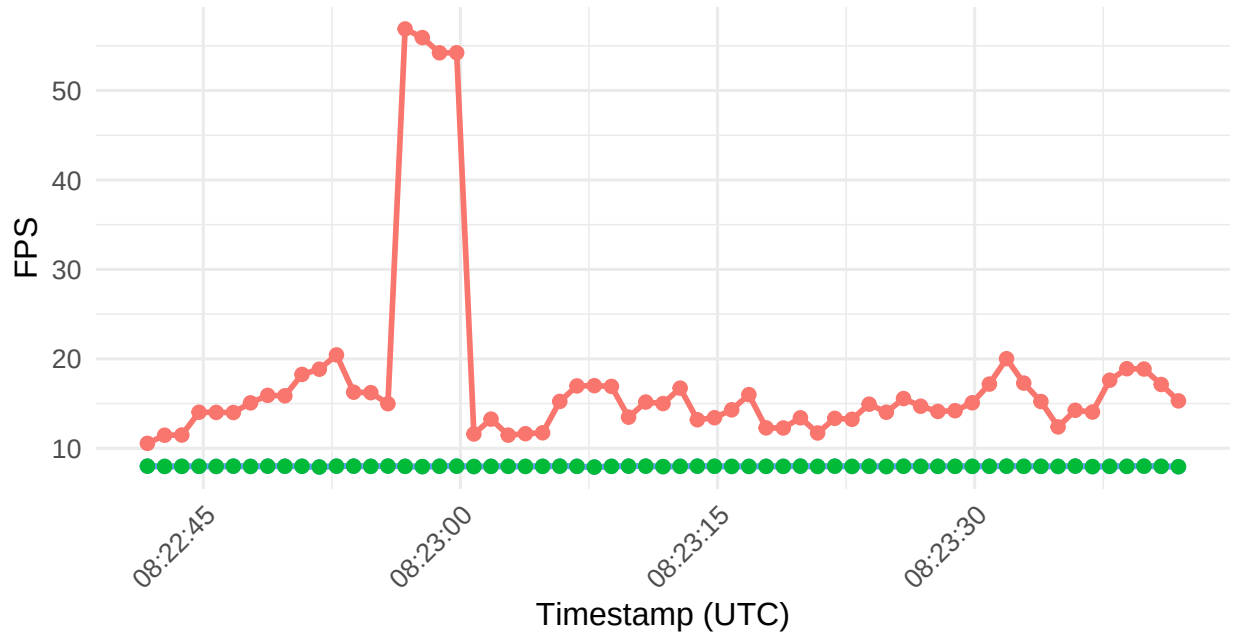
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 8
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: attempting to establish new baseline on t420 with dedicated camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_sr8.csv")
```

Performance Monitoring Metrics

File: fps_t420_logitech_sr8.csv



Metric —●— Camera_FPS —●— Inference_FPS —●— Sampling_FPS

Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

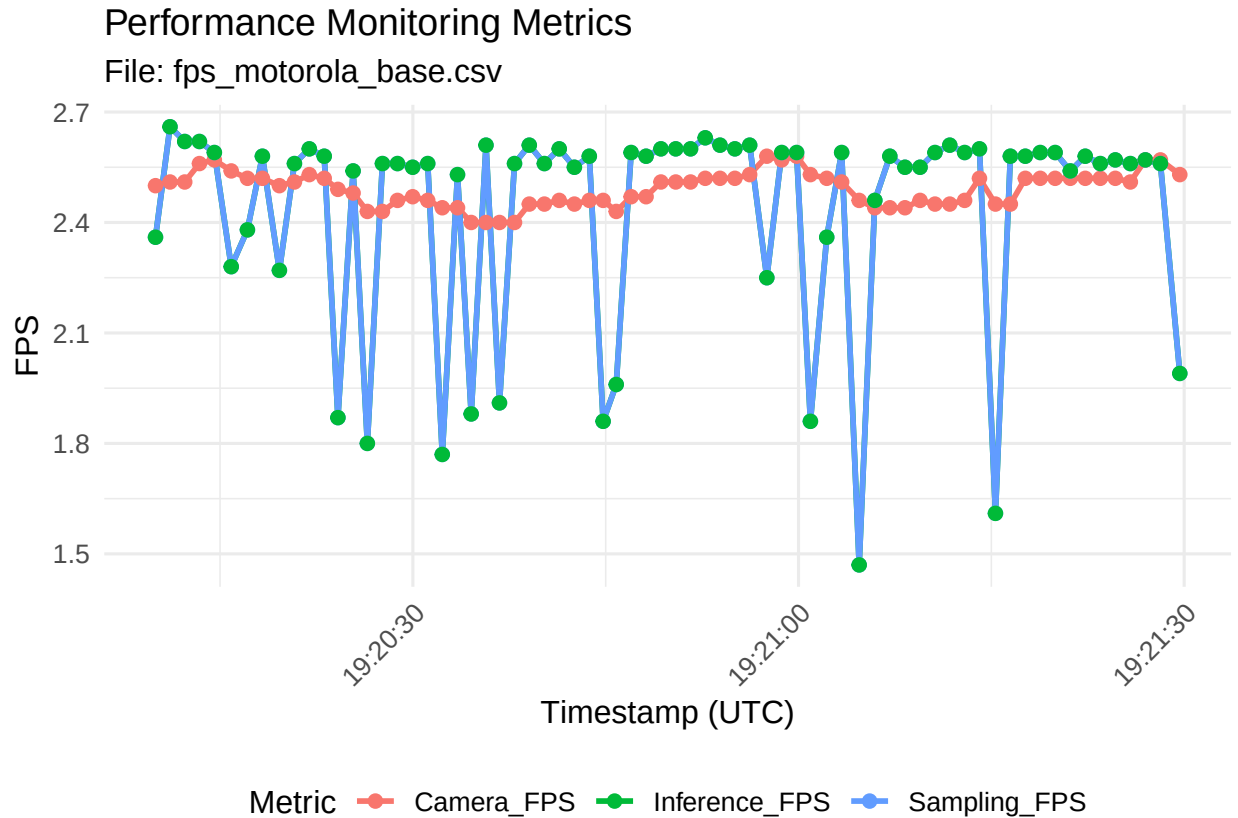
Notes: Camera_FPS continues to act weird, with a huge 50+ fps peak lasting a few seconds. Baselinewise: a sr8 keeps Inference_FPS and Sampling_fps at a constant 8, always below the camera curve. We now consider sr8 an acceptable baseline for t420 + logitech. Next up: measurements on a mobile device.

Test 2.8 Running on mobile

Environment

- device: Motorola moto g06 (MediaTek Heli G81 Extreme, 4GB+2GB(RAM boost), no dedicated GPU)
- OS: Android 15
- browser + version: Mozilla/5.0 (Android 15; Mobile; rv:157.0) Gecko/157.0 Firefox/157.0
- camera: Front 8M
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: Attempting to run application on mobile
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_motorola_base.csv")
```



Results

Can't access browser command line on mobile.

Notes: On mobile we're shockingly far away from our targets. **Camera_FPS** is between 2 and 3, while inference and sampling are between 1 and 3. I will decrease the target to 2, and then try to find a better baseline.

Test 2.9 Baseline on mobile

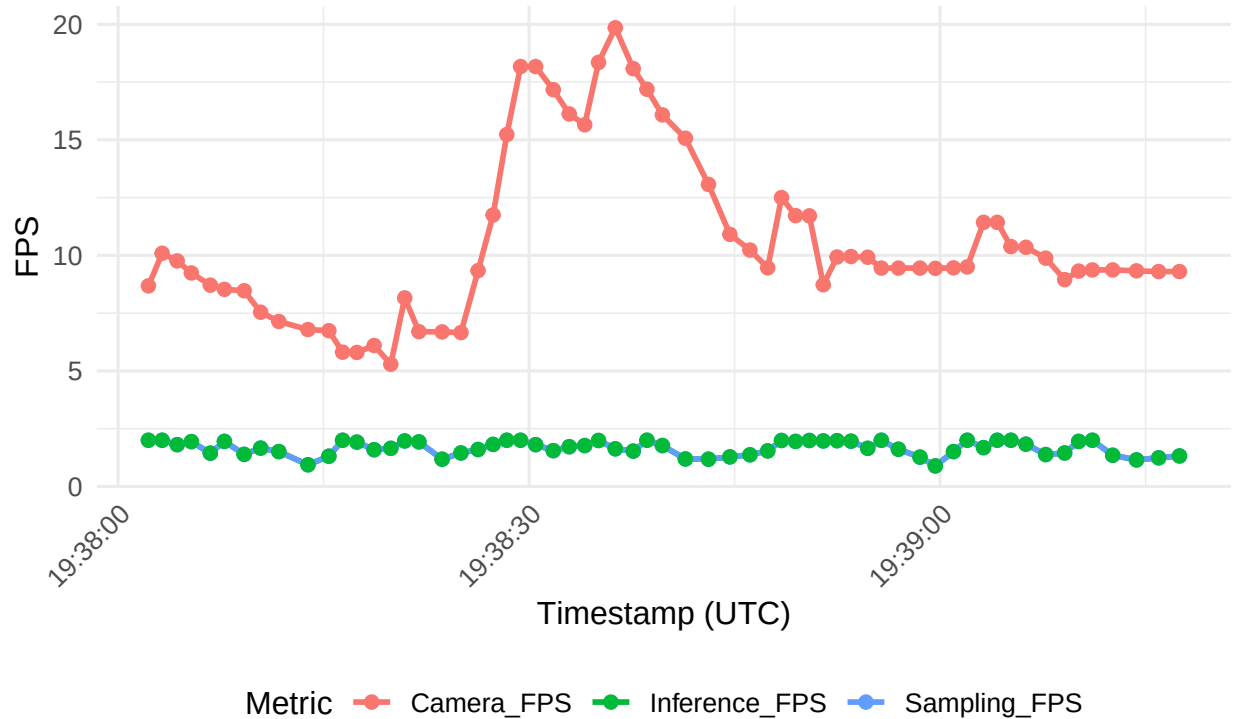
Environment

- device: Motorola moto g06 (MediaTek Heli G81 Extreme, 4GB+2GB(RAM boost), no dedicated GPU)
- OS: Android 15
- browser + version: Mozilla/5.0 (Android 15; Mobile; rv:157.0) Gecko/157.0 Firefox/157.0
- camera: Front 8M
- target sampling rate: 2
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: Attempting to find a baseline on mobile
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_motorola_sr2.csv")
```

Performance Monitoring Metrics

File: fps_motorola_sr2.csv



Results

Can't access browser command line on mobile.

Notes: Changing the SR target to 2 freed up the camera FPS. Now camera FPS moves between 5 and 20. Sampling & inference can't reach 2 in a stable fashion (multiple dips below). Nevertheless I think that it is still worth it to attempt SR3, possibly even higher if SR3 goes well.

Test 2.10 Baseline on mobile: SR3

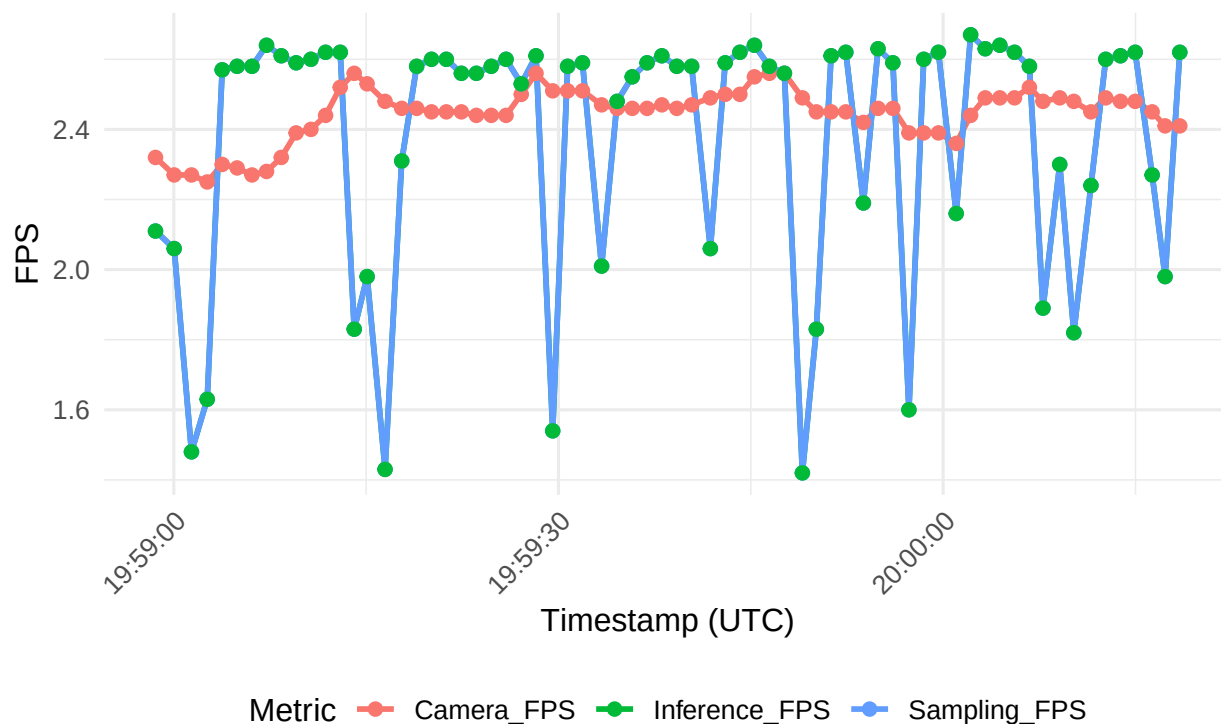
Environment

- device: Motorola moto g06 (MediaTek Heli G81 Extreme, 4GB+2GB(RAM boost), no dedicated GPU)
- OS: Android 15
- browser + version: Mozilla/5.0 (Android 15; Mobile; rv:157.0) Gecko/157.0 Firefox/157.0
- camera: Front 8M
- target sampling rate: 3
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: Attempting to find a baseline on mobile
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_motorola_sr3.csv")
```

Performance Monitoring Metrics

File: fps_motorola_sr3.csv



Results

Can't access browser command line on mobile.

Notes: We already hit the expected problems (trying to sample at a higher rate than what our latency allows for thus limiting the **Camera_FPS** and seldom reaching our sampling rate + causing wild fluctuations. Therefore we establish 2 as a mobile sample rate baseline.